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Differential behavior of *Lactobacillus helveticus* B1929 and ATCC 15009 on the hydrolysis and angiotensin-I-converting enzyme inhibition activity of fermented ultra-high-temperature milk and nonfat dried milk powder

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ABSTRACT

Consumers' growing interest in fermented dairy foods necessitates research on a wide array of lactic acid bacterial strains to be explored and used. This study aimed to investigate the differences in the proteolytic capacity of *Lactobacillus helveticus* strains B1929 and ATCC 15009 on the fermentation of commercial ultra-pasteurized (UHT) skim milk and reconstituted nonfat dried milk powder (at a comparable protein concentration, 4%). The antihypertensive properties of the fermented milk, measured by angiotensin-I-converting enzyme inhibitory (ACE-I) activity, were compared. The B1929 strain lowered the pH of the milk to 4.13 ± 0.09 at 37°C after 24 h, whereas ATCC 15009 needed 48 h to drop the pH to 4.70 ± 0.18 at 37°C. Two soluble protein fractions, one (CFS1) obtained after fermentation (acidic conditions) and the other (CFS2) after the neutralization (pH 6.70) of the pellet from CFS1 separation, were analyzed for D-/L-lactic acid production, protein concentration, the degree of protein hydrolysis, and ACE-I activity. The CFS1 fractions, dominated by whey proteins, demonstrated a greater degree of protein hydrolysis (7.9%) than CFS2. On the other hand, CFS2, mainly casein proteins, showed a higher level of ACE-I activity (33.8%) than CFS1. Significant differences were also found in the D- and L-lactic acid produced by the UHT milk between the 2 strains. These results attest that milk casein proteins possessed more detectable ACE-I activity than whey fractions, even without a measurable degree of hydrolysis. Findings from this study suggest that careful consideration must be given when selecting the bacterial strain and milk substrate for fermentation.

Key words: *Lactobacillus helveticus*, ultra-pasteurized milk, nonfat dried milk, fermentation, ACE inhibition

INTRODUCTION

Fermented foods are "foods or beverages made through controlled microbial growth and the conversion of food components through enzymatic conversion of major and minor food components" (p. 94, Marco et al., 2017). Fermentation practices date back as much as 13,000 yr (Liu et al., 2018; Mannaa et al., 2021). They were used by early civilizations for food preservation by improving the safety, digestibility, and organoleptic properties of foods (González et al., 2019). Furthermore, historical records show fermentation of several substrates, including milk and cereal, is indigenous to many parts of the world (Mannaa et al., 2021). Many foods can undergo fermentation, including meat and fish, dairy, vegetables, soybeans, other legumes, cereals, and fruits (Dimidi et al., 2019). In recent years, fermented foods have experienced a surge in popularity (Technavio, 2022), mainly due to their proposed nutritional and health benefits, including fermented dairy foods.

Fermented dairy products provide a variety of nutrients, including fatty acids, vitamins, and minerals. They may also contain bioactive peptides, prebiotic substances, and probiotic bacteria, providing additional health benefits, including disease prevention and treatment (Nicoletti, 2012; González et al., 2019). Epidemiological studies have shown an association between consuming fermented dairy foods, weight management, and reducing the risk of cardiovascular disease, metabolic disorders, and immune-related pathologies (González et al., 2019). Milk proteins, especially caseins, can be hydrolyzed, releasing bioactive peptides (Giacometti Cavalheiro et al., 2020) that can positively affect the physiological conditions of the body beyond their known nutritional value (Korhonen and Pihlanto, 2006). These peptides can regulate essential bodily

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functions through antihypertensive, antimicrobial, antithrombotic, immunomodulatory, opioid, antioxidant, and mineral binding processes (Chakrabarti et al., 2018).

Lactic acid bacteria (**LAB**) are auxotrophic for many EAA, requiring their uptake from the surrounding environment. To do this, many LAB have evolved highly sophisticated proteolysis and peptide-uptake systems to survive within the dairy environment, producing a large and diverse pool of bioactive peptides (Raveschot et al., 2018; Kieliszek et al., 2021). The LAB proteolytic system comprises cellular envelope proteases (**CEP**) that initially hydrolyze proteomes, transporters that import peptides, and intracellular peptidases that further hydrolyze peptides (Raveschot et al., 2018). One of the most well-studied biological activities of fermented dairy peptides is the inhibition of angiotensin-converting enzyme (**ACE**) activity (Boelsma and Kloek, 2009). Angiotensin-converting enzyme is a part of the renin-angiotensin system and regulates blood pressure (Patel et al., 2016). The activation of ACE converts angiotensin I to angiotensin II, a vasoconstrictor, leading to elevated blood pressure (Su et al., 2021). Hypertension affects over 1 billion adults worldwide (World Health Organization, 2021). Foods or ingredients that can potentially mitigate this condition would bear a broad effect. Lactic acid bacteria-fermented milk is one example of a product that can help lower the blood pressure of hypertensive subjects (Pripp, 2008; Chen et al., 2021).

Among LAB, *Lactobacillus helveticus* in particular has a unique and extensive proteolytic system (Sadat-Mekmene et al., 2011; Raveschot et al., 2018). In a survey of several starter cultures in sterilized skim milk, including *Lactobacillus acidophilus*, *Lacticaseibacillus casei*, and *Lactobacillus delbrueckii*, the *L. helveticus* strain reportedly displayed the highest proteolytic activity and highest ACE-inhibiting activity (El-Fattah et al., 2016). A study comparing *L. helveticus* with *Limosilactobacillus reuteri* and *Lacticaseibacillus rhamnosus* also evidenced that *L. helveticus* showed the most significant proteolytic activity (Begunova et al., 2021). This is mainly due to many strains possessing 1 or more homologs of CEP gene *prtH* in their genomes, many of which can contain more than 1 or up to 4 (Broadbent et al., 2011; Nejati et al., 2016). However, various combinations of these CEP genes (Nejati et al., 2016) also lead to diverse bioactivities depending on the strains. For example, certain strains produce acid in milk at varying rates, highlighting their diversity of metabolic activities (Fortina et al., 1998). *Lactobacillus helveticus* ATCC 15009, which contains the CEP gene *prtH3* (Broadbent et al., 2011), was classified as a medium-acidifying strain compared with those that lowered the pH more quickly or slowly. The consumption of fermented milk containing specific *L. helveticus* strains can help reduce blood pressure in patients suffering from hypertension (Hata et al., 1996; Seppo et al., 2003). The ACE-inhibiting potential of the *L. helveticus* proteomes remains to be explored and effectively used to develop a wide range of probiotic dairy foods.

In addition to the importance of strain selection, the quality of the dairy growth media should be considered when used to produce bioactive components. Heat is often employed to ensure the safety and uniformity of fermented dairy products. Heating temperature and holding times affect the final product textures due to irreversible conformational changes to milk proteins (Sfakianakis and Tzia, 2014). Different heat treatments of milk can also affect final growth and pH after inoculation with LAB (O'Toole, 1995). *Lactococcus lactis* strains reportedly produced more significant amounts of acid when grown in milk heated to 115°C versus 80°C (O'Toole, 1995; Laligant et al., 2003), indicating that heat treatment of milk can affect the downstream fermentation process. Even heating milk to 82°C versus 90°C can have differing effects of the final titratable acidity of fermented milk inoculated with starter cultures (Greene and Jezeski, 1957). Furthermore, raw, pasteurized, or ultra-pasteurized (UHT) skim milk fermented by *L. acidophilus*, *L. delbrueckii*, *L. casei*, *Limosilactobacillus fermentum*, or *L. rhamnosus* strains exhibited differential protein hydrolysis after 24 h (Parrella et al., 2012). Several fermentation studies (Chen et al., 2014; Begunova et al., 2021; Loghman et al., 2022) used pre-heated reconstituted NDM powder or commercially pasteurized milk. However, it is not known whether pre-heating treatment favor or hinder the growth and proteolytic activity of specific *L. helveticus* strains such as those that acidify milk at different rates.

We report our initial findings on the differential behavior of *L. helveticus* B1929 and ATCC 15009 strains in fermenting reconstituted commercial NDM powder and UHT pasteurized skim milk to understand the interaction effects of bacterial strain selection and milk pre-treatment on the ACE inhibition activities of the fermented products.

MATERIALS AND METHODS

Materials

No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board. Di-sodium tetraborate decahydrate, SDS (99%), o-phthaldialdehyde (**OPA**), dithiothreitol (99%), L-serine (Ser), furanacryloyl–prolyl-glycyl-glycine, Tris-HCl,

sodium chloride (NaCl), and angiotensin-I-converting enzyme from rabbit lung (ACE, EC 3.4.15.1) were purchased from Sigma Aldrich Co.

Organic NDM powder was a gift from Milk Specialties Global (PIN 100696, Eden Prairie, MN). The protein content of the NDM, as specified by the supplier, is ≥32.0% (by N analysis, as-is), fat ≤1.5%, ash ≤8.0%, and aerobic plate count (**APC**; cfu/g) ≤ 10,000. Organic ultra-pasteurized (UHT) nonfat milk was procured from a local grocery store. Ultra-high-temperature pasteurization refers to heating the milk to above 138°C for at least 2 s in accordance with the US Food and Drug Administration regulations (21CFR131.3). The protein content of the UHT milk is 4.0%, as reported by the manufacturer.

Heat Treatment of Reconstituted NDM

The heat treatment of NDM (Sfakianakis and Tzia, 2014) was performed after adding 20.6 g of NDM powder to 200 mL of sterilized distilled water and mixing thoroughly with stirring until fully suspended. The reconstituted milk solution (effective protein content ~4.0%) was heated to 90 to 95°C for 5 min and then immediately placed on ice (**NDM95**). Another solution was prepared similarly, but the heating temperature and duration were adjusted to 85 to 90°C and 20 min (**NDM85**), respectively.

Bacterial Strains

Lactobacillus helveticus B1929 and ATCC 15009 (B5426) were from the in-house collection (Agricultural Research Service Culture Collection). Cultures were streaked onto de Man, Rogosa, and Sharpe agar (Difco) from 15% (vol/vol) glycerol stocks stored at −70°C and incubated at 33°C for 72 h. Single colonies were inoculated into 5 mL of de Man, Rogosa, and Sharpe broth and incubated statically at 37°C for 24 h and room temperature for 2 min. Overnight cultures were centrifuged at 15,000 × *g*, rinsed in 0.1% (wt/vol) peptone water (Difco) twice, and finally resuspended in equal volumes of NDM95 milk.

DNA Extraction and PCR

Genomic DNA was extracted from overnight *L. helveticus* cultures using the DNeasy Blood and Tissue Kit (Qiagen). Individual endpoint PCR reactions were then carried out for CEP genes using primers previously designed to target *prtH* (F 5′-CGGGAATCCTGATATGGAAACTTTAGGTAATC; R 5′-ATTAAGACTAGCAAAGATTGGAAGATCTGAACTTG) (Pederson et al., 1999), *prtH2* (F 5′-AAGCAAAGGATGTTGTTCCAAGTAAGCCA; R 5′-CTCTCTTCCTTCTTACCAGTTGATGATTGAACT) (Smeianov et al., 2007), *prtH3* (F 5′-GATGATCAAGCAGATGTAAAACCGGCAGAAG; R 5′-ATTTACTGAAGAATTAGTCAAATGACCTGTTGTCGG) (Broadbent et al., 2011), and *prtH4* (F 5′-CTGAAGCAGCAACTAATGATCCTGG; R 5′-TGGATTAGGATCCGTTCTGGTTGTCAG) (Broadbent et al., 2011), where F = forward and R = reverse. Sizes were 1,332, 4,042, 357, and 3,386 bp, respectively. Each 25-μL reaction contained 12.5 μL of GoTaq Green Master Mix (Promega), 2.5 μL each of 10 μ*M* primer, 1 μL of template DNA, and 6.5 μL of nuclease-free water (Qiagen). The first cycle was set at 95°C for 2 min, followed by 30 cycles at 95°C for 30 s, 54°C for 30 s, then 65°C for 3 min, and lastly, an extension at 65°C for 3 min (Broadbent et al., 2011). The PCR products were visualized using a 1% agarose gel.

Bacterial Fermentation and Sample Preparation

Lactobacillus helveticus was inoculated in 10 mL of UHT, NDM95, or NDM85 milk (in triplicate) at a starting concentration of 5.5 to 5.8 $\log_{10}$ cfu/mL. Samples were incubated aerobically without shaking for 24 h (B1929) or 48 h (ATCC 15009) at 37°C. Uninoculated 10 mL of milk samples (negative controls) were also incubated under the same conditions for the same lengths of time. Enumeration was performed by 10-fold serial dilutions with 0.1% (wt/vol) peptone water, and 100 μL were spread in triplicate on de Man, Rogosa, and Sharpe agar plates. Plates were incubated at 37°C for 2 d before counting. Only colony morphologies matching that of either *L. helveticus* strain were counted for APC.

The pH values of all inoculated samples and negative controls were measured using an IQ240 pH meter (IQ Scientific Instruments). We used DL-lactic acid (1.0 *M*, J. T. Baker) to decrease the negative control samples' pH (±0.05) to their corresponding inoculated counterparts, designated as UHT-AS, NDM95-AS, or NDM85-AS. For each sample, 7 mL was centrifuged at 15,000 × *g* at 4°C for 30 min. The supernatant was filtered with a syringe-driven filter (0.22-μm low protein-binding PVDF, MilliporeSigma), designated as cell-free supernatant 1 (**CFS1**). The remaining pellet was resuspended in 6 mL of sterile deionized (**DI**) water, and the pH was carefully adjusted to ~6.70 using 1 *M* NaOH. This neutralized fraction was again centrifuged at 15,000 × *g* at 4°C for 30 min, and the supernatant was filtered with a syringe-driven filter (0.22-μm low protein-binding PVDF, MilliporeSigma), designated as CFS2. The negative controls of CFS2 were designated as UP-NS, NDM95-NS, or

NDM85-NS. The mass of the remaining pellet was recorded. All cell-free supernatants, CFS1 and CFS2, were stored at 4°C until analysis.

Lactic Acid Assay

We quantified D- and L-lactic acid concentrations using the Megazyme D-/L-Lactic Acid Assay Kit (K-DLATE, Megazyme International Ireland Ltd.) according to the manufacturer's protocol. These concentrations were determined by measuring the increase of NADH using an Eppendorf BioPhotometer plus (Eppendorf) at 340 nm.

Gel Electrophoresis

Pre-cast NuPAGE 12%, Bis-Tris, 1.0-mm mini protein gels (10 wells, Invitrogen) were used to carry out SDS-PAGE (Xiao, et al., 2020). One representative CFS1 and CFS2 sample (from the triplicates) for each B1929, ATCC 15009, and the negative controls were prepared for the gel analysis following the manufacturer's instructions. NuPAGE Sample Reducing Agent (10×; Invitrogen) containing 500 m*M* dithiothreitol and NuPAGE Antioxidant (Invitrogen) was used for all samples. Cell-free supernatant 1 (2.5 μL) and CFS2 (2.5 μL) were mixed with the NuPAGE LDS Sample Buffer (4×; Invitrogen) and DI water to add up to a total of 10 μL for each sample. Gels were run using NuPAGE MES SDS Running Buffer (20×; Invitrogen) in an XCell SureLock Mini-Cell (Invitrogen) at 200 V for 50 min. The Precision Plus Protein Dual Color Standards (Bio-Rad) were also loaded (10 μL). The gels were then rinsed thoroughly 3 times for 5 min with DI water, then stained for 1 h with SimplyBlue SafeStain (Invitrogen), followed by de-staining in DI water twice for 1 h each.

Estimation of Protein Concentration

The protein concentration of each fraction, CFS1 and CFS2, from the various fermented milk was estimated by measuring the absorbance at 278 and 280 nm, respectively (Cary 60 UV-VIS spectrometer, Agilent Technologies). Any light scattering at 330 nm was corrected when necessary (Grimsley and Pace, 2003). The extinction coefficient $\varepsilon_{278} = 1.046$ (Mahmoudi et al., 2007) and $\varepsilon_{280} = 1.00$ mL/cm/mg (Creamer et al., 1982) was used for CFS1 and CFS2, respectively. Appropriate dilutions (10–50×) were made for each sample using Milli-Q water to obtain the absorbance values within the linear range of the Beer-Lambert law (absorbance within 0.5–1.2).

Determination of Degree of Protein Hydrolysis

The degree of protein hydrolysis (**DH**) of each soluble fraction, CFS1 and CFS2, was determined using the previously published OPA method (Nielsen et al., 2001) with some modifications. Briefly, 200 mL of 0.1 *M* tetraborate buffer (pH 9.2) containing 3.0 m*M* SDS, 5.7 m*M* dithiothreitol, and 6.0 m*M* OPA (from a 0.3 *M* stock solution in 4.0 mL ethanol) was prepared. The serine standard solution (0.9516 mEq/L) was made with Milli-Q water. One hundred microliters of diluted (6 to 12 times in Milli-Q water) sample solution (CFS1 and CFS2) were mixed with 3.0 mL of OPA and incubated for 2 min at room temperature (20°C) in each assay preparation. The spectrophotometric reading was set at 340 nm against Milli-Q water as the reference. The serine standard (400 mL) was also prepared and measured similarly (in 3.0 mL of OPA solution). Two sets of blank (400 mL of Milli-Q water and 3.0 mL of OPA solution), serine standard, and sample solutions were performed in each DH determination. The protein concentration was used as estimated spectrophotometrically (described above). The DH value was calculated as described by Nielsen et al. (2001) using the α, β, and h_{tot} values of 1.0, 0.40, 8.8, and 1.04, 0.38, 8.2 for CFS1 and CFS2, respectively, adapted from Adler-Nissen (Adler-Nissen, 1986) using the following equations:

$$\text{Ser-NH2} = [(\text{OD}_{\text{sample}} - \text{OD}_{\text{blank}})/(\text{OD}_{\text{standard}} - \text{OD}_{\text{blank}})] \times 0.9516 \times \text{L}/(\text{X} \times \text{P}), \quad [1]$$

$$\text{h} = (\text{Ser-NH2} - \beta)/\alpha, \quad [2]$$

$$\text{DH}\ (\%) = (\text{h}/\text{h}_{\text{tot}}) \times 100\%, \quad [3]$$

where Ser-NH2 is the Ser-NH2 (mEq/L)/g protein, P is the protein concentration (%) in the sample, L is the sample volume in liters (= 0.1 L), X is the sample weight (~100 g in 0.1 L in this work), and h_{tot} is the total number of peptide bonds in the protein.

Angiotensin-I-Converting Inhibition Activity

The angiotensin-I-converting inhibition activity (**ACE-I**) inhibitory activity of CFS1 and CFS2 was assessed using the reported furanacryloyl–prolyl-glycyl-glycine based substrate assay procedure (Shalaby et al., 2006) with some modifications. The final substrate (furanacryloyl–prolyl-glycyl-glycine) concentration was 0.88 m*M* in 50 m*M* Tris-HCl buffer (pH 7.5) with 0.3 *M* NaCl. The ACE solution (0.25 units/mL) was prepared fresh each day by adding to Milli-Q water and

mixing thoroughly. The assay was performed using 5× and 20× diluted CFS1 and CFS2 sample solutions. The reaction started (at 37°C) with a Peltier temperature controller (Quantum Northwest) with gentle stirring as soon as the substrate (pre-heated at 37°C for 15 min) was added to the sample and buffer mixture. The absorbance at 340 nm was recorded every 30 s for 20 min. The slope averaged over a linear range up to 15 min was taken as a measurement of the ACE-I activity. The ACE-I was calculated using the following equation:

$$\text{ACE-I}\ (\%) = [1 - (\Delta A_{\text{inhibitor}}/\Delta A_{\text{control}})] \times 100, \quad [4]$$

where $\Delta A_{\text{inhibitor}}$ and $\Delta A_{\text{control}}$ are the slopes of the sample with inhibitor and control, respectively, all assays were performed in triplicates.

Fermentate Mixture Clean-Up

Five hundred microliters of CFS1 sample (from UP milk fermentation) were passed through an Amicon Ultra-0.5 Centrifugal Filter Unit (MW 10-kDa cut-off; MilliporeSigma). Before use, the filter unit was rinsed with 50% methanol twice and followed by DI water. The filtrate was concentrated in a CentriVap Benchtop Vacuum Concentrator (Labconco) at 37°C. A 0.5% formic acid solution (10 mL) was mixed with the concentrated CFS1 sample (40 mL) to reach the final concentration of 0.1% formic acid (per manufacturer's instructions). A 100-mL capacity C18 Omix tip (Agilent) pre-equilibrated with 0.1% formic acid was used to bind and clean up the CFS1 sample. The bound peptides were eluted using a washing solution of 0.1% formic acid in 50% acetonitrile.

Peptide Identification by Liquid Chromatography-MS

Liquid chromatography-MS (**LC-MS**) analyses were performed using a Nano Acuity (Waters) system connected to a QE+ Orbitrap (Thermo Fisher Scientific) mass spectrometer. Two microliters (2.0 μL) of the sample were injected into a diphenyl column (1,000 Å 2.7 μm, 0.3 × 100 mm; HALO) at a flowrate of 0.01 mL/min and temperature of 40°C. Mass spectrometry grade methanol, acetonitrile, and formic acid (MilliporeSigma) were used. Mobile phase (A) was 0.1% formic acid in ultra-pure water, and a mobile phase (B) was 0.1% formic acid in 90% acetonitrile in DI water. A gradient flow was set starting at 1% B, ramping to 100% B over 40 min, keeping at 100% B for 30 min, then ramping to 1% B for 20 min (total run time of 90 min). Ionization was performed using a heated electrospray probe on the Ion Max source in positive mode. The ion source temperature was 54°C, capillary temperature of 300°C, and radio frequency lens voltage of 70 V. A cone voltage of 3.0 kV, auxiliary gas of 3 arbitrary units, and sheath gas of 7 arbitrary units (no sweep gas) were used.

Proteomic data were analyzed with Proteome Discoverer version 2.3 (Thermo Fisher Scientific). The bovine database was downloaded from UniProt and uploaded into Proteome Discoverer for analysis.

Statistical Analysis

All statistical analyses were performed using Microsoft Excel and R 4.1.1 (R Core Team, 2021). Figures were generated using R packages ggplot2 (Wickham and Chang, 2016) and connecting letters were identified using multcompView (Graves et al., 2019). Normality for each data set was assessed by examining the q-q plots of the residuals. Two-way ANOVA with interaction was used to analyze the $\log_{10}$-transformed APC and the final pH. Tukey's honestly significant difference test was performed for multiple pairwise comparisons. Two-way ANOVA with interaction was also conducted for CFS1 and CFS2 metrics for *L. helveticus*-fermented samples only, and Tukey's honestly significant difference was used for multiple pairwise comparisons. Pearson correlation coefficients were calculated to determine the linear relationships between data sets. A positive correlation was specified to be $\rho > 0.60$, whereas a negative correlation was specified to be $\rho < -0.60$, and a value $-0.60 < \rho < 0.60$ was not considered to be correlated. The significance cut-off was defined as $\alpha < 0.05$.

RESULTS AND DISCUSSION

Selection of L. helveticus Strains

Lactobacillus helveticus B1929 and ATCC 15009 strains were selected for this study based on a preliminary PCR screening from which B1929 was identified as having 2 CEP genes, *prtH3* and *prtH4*, whereas ATCC15009 contained only *prtH3*. These results agreed with a previous survey that *L. helveticus* CEP gene profiles were strain-specific (Broadbent et al., 2011). The difference in CEP gene profiles may contribute to differing abilities to hydrolyze proteins and, ultimately, release bioactive peptides.

The 2 *L. helveticus* strains were examined for their ability to grow and ferment skim UHT milk and pre-heated reconstituted NDM. Both strains were inoculated at an initial concentration between 5.5 to 5.8 $\log_{10}$ cfu/mL and were able to grow and coagulate the milk, but to differing extents. Figure 1 shows the differences

in APC and pH at the end of the fermentation. The B1929 strain grew in all milk media to an average of 8.97 ± 0.03 $\log_{10}$ cfu/mL, and generally better than ATCC 15009 in UHT, NDM95, and NDM85 by 1.37 $\log_{10}$ cfu/mL ($P < 0.001$), 1.07 $\log_{10}$ cfu/mL ($P < 0.001$), and 2.19 $\log_{10}$ cfu/mL ($P < 0.001$), respectively. Additionally, ATCC 15009 in NDM95 and NDM85 grew greater than in UHT by 1.14 $\log_{10}$ cfu/mL ($P < 0.001$) and 1.00 $\log_{10}$ cfu/mL ($P < 0.001$), respectively. In another study (Nejati et al., 2016) also using reconstituted NDM, all *L. helveticus* strains were able to grow to above 8.3 $\log_{10}$ cfu/mL, nearly identical to the 2 strains in both pre-heated NDM media. However, the slower growth of ATCC 15009 in UHT in this study was highly notable.

Effect of L. helveticus Strains on Lactic Acid Production

Even though B1929 was only a medium-acidifying strain (Fortina et al., 1998), the total APC grew to above 9 $\log_{10}$ cfu/mL (Figure 1A) in all 3 milk media. This APC was slightly higher than many of the 24-h fast-acidifying strains used by Nejati et al. (2016), where the incubation temperature was 40°C, also slightly elevated compared with the 37°C used in this work. A study (Begunova et al., 2021) using a fast-acidifying *L. helveticus* strain at a higher starting concentration of cells grew to above 9 $\log_{10}$ cfu/mL at 37°C.

The B1929 strain lowered the pH to an average of 4.13 ± 0.09 among all fermented milk samples after 24 h at 37°C (Figure 1A). The pH of the milk fermented by ATCC 15009 only lowered to 5.82 ± 0.02 after the same amount of fermentation time, suggesting it should be classified as a slow-acidifying strain, even though it had been classified as a medium-acidifying strain previously (Fortina et al., 1998). After 48 h of incubation, the average pH reached 4.70 ± 0.18, significantly higher ($P < 0.001$) than B1929 fermentates at 24 h. Furthermore, the pH of the ATCC 15009 UHT fermentate was significantly greater ($P = 0.028$), 0.62 unit, than the fermentates recovered from its growth in NDM and the growth of B1929 in all milks.

Figure 1B shows the concentrations of D- and L-lactic acid enantiomers. The B1929 strain produced 55.6 m*M* more total lactic acid than ATCC 15009 in UHT ($P < 0.001$). This was driven by greater production of both D-lactic acid (23.9 m*M*, $P < 0.001$) and L-lactic acid (31.7 m*M*, $P = 0.038$) in UHT. We found no significant differences in total, D-, or L-lactic acid production between the 2 strains in NDM, except that B1929 produced more D-lactic acid in NDM85 than ATCC 15009

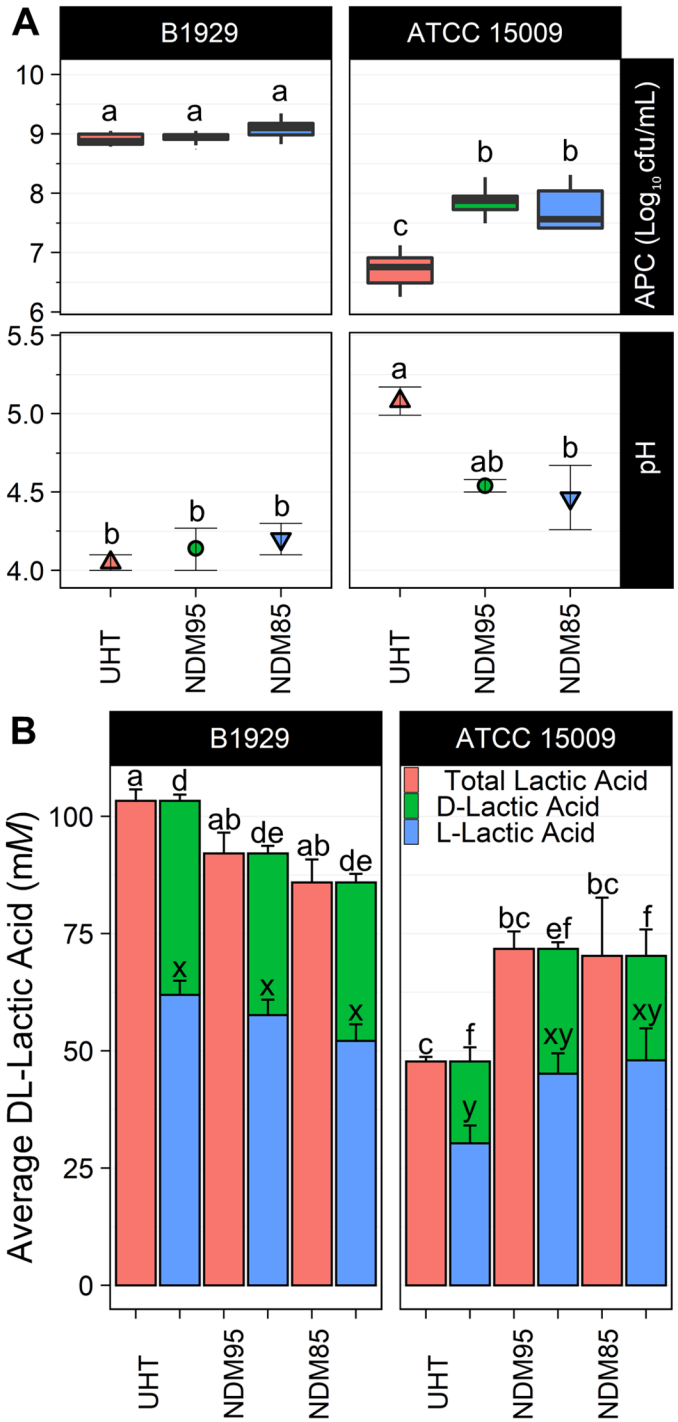


Figure 1. (A) Aerobic plate counts (APC, boxplots) and final pH of *Lactobacillus helveticus* B1929 and ATCC 15009 (shapes) in each milk medium. Data sharing the same letters were not significantly different ($P > 0.05$). Means (n = 3) and SE bars indicated for pH. (B) The total, D-lactic acid, and L-lactic acid concentrations (m*M*) were produced by *L. helveticus* B1929 and ATCC 15009. NDM95 = reconstituted milk solution heated to 90 to 95°C for 5 min; NDM85 = reconstituted milk solution heated to 85 to 90°C for 20 min.

by 11.6 mM ($P = 0.049$). These results supported that *L. helveticus* produces a racemic mixture of D- and L-lactic acids (Aeschlimann and von Stockar, 1991). D-Lactic acid was generally thought to be slowly metabolized in mammals, but growing evidence revealed its substantial metabolism in humans (Monroe et al., 2019).

Effect of L. helveticus Strains on Protein Composition

Figures 2A and 2B show the protein compositional profiles of CFS1 in the milk fermented by B1929 and ATCC 15009. These gel patterns attested that CFS1 was dominated by whey proteins, especially in the fermented samples (lanes 2, 4, and 6). This is more pronounced by the B1929 strain fermentation (Figure 2A). The difference between fermented samples and negative controls (i.e., less casein present in the fermented samples) can be attributable to the proteolysis that occurred during fermentation. Two unique bands, significantly smaller than 10 kDa, appeared in all fermented CFS1 samples (lanes 2, 4, and 6). The band with the smallest molecular weight (indicated with a box) was more prominent and, thus, deserved to be further investigated by LC-MS for its possible association with the ACE-I activity observed in this work.

Casein was evidently detected in nearly all negative controls of CFS1 (uninoculated and acidified to comparable pH values; lanes 1, 3, and 5) and was more noticeable in the CFS1 of UHT and NDM85 (lanes 2 and 6; Figure 2B) fermented by ATCC 15009 than by B1929 strain despite their lowered acidity (Figure 1), pH = 4.94 ± 0.18 versus 4.05 ± 0.05. This final higher average pH value alone could not explain why more casein was present as it was also closer to the pI of whey protein (~4.8; Pelegrine and Gasparetto, 2005). Additionally, all negative controls obtained at these 2 pH conditions (Figures 2A and 2B) showed similar protein composition profiles. Therefore, we assumed the difference in the proteolysis patterns of the 2 strains was accountable for the distinction in the protein composition.

The CFS1 of NDM95 by ATCC 15009 showed a less detectable level of casein (lane 4; Figure 2B) than UHT and NDM85, suggesting higher temperature (95°C) preheating NDM altered the protein uptake behavior by ATCC 15009 and proteolysis pattern.

Figures 3A and 3B show the protein compositional profiles of CFS2 in the milk samples fermented by B1929 and ATCC 15009. All fractions were obtained by neutralizing the coagulated pellet to pH ~6.70, leading to the re-solubilization of casein. These gels corroborated that casein was the predominant component in these

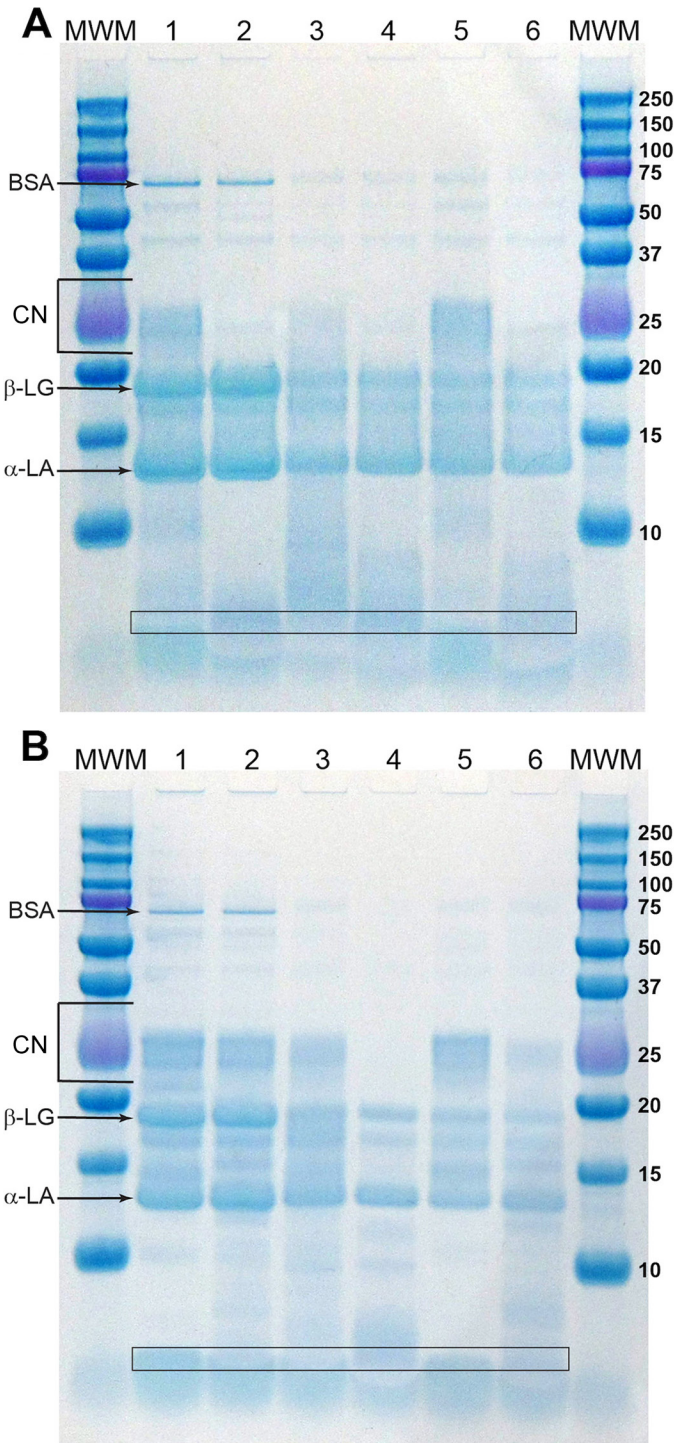


Figure 2. Sodium dodecyl sulfate-PAGE of cell-free supernatant 1 from UHT, reconstituted milk solution heated to 90 to 95°C for 5 min (NDM95) or 85 to 90°C for 20 min (NDM85) fermented by *Lactobacillus helveticus* (A) B1929 and (B) ATCC 15009. The lanes were loaded as follows: (1) negative control UHT, (2) *L. helveticus* UHT, (3) negative control NDM95, (4) *L. helveticus* NDM95, (5) negative control NDM85, and (6) *L. helveticus* NFDM85. Molecular weight markers (MWM) are also indicated. Unique and yet to be identified bands (due to fermentation) are marked in boxes.

samples. As expected, all negative controls exhibited similar protein composition (lanes 1, 3, and 5). Similar to CFS1 (Figure 2), 2 unique bands, significantly smaller than 10 kDa (the lowest molecular marker), were visible in all fermented CFS2 samples (lanes 2, 4, and 6). As with CFS1 (Figure 2), the smallest band was further studied by LC-MS (discussed below).

Much like its counterpart (CFS1, discussed above), the CFS2 of NDM95 by ATCC 15009 strain also evidenced the presence of various smaller molecular weight species and the absence of intact casein proteins (lane 4; Figure 3B), further attesting likely additional protein damage due to the pre-heating treatment.

All 3 kinds of milk, UHT, NDM95, and NDM85, displayed noticeable differences in their protein composition in the CFS1 and CFS2. The differences between UHT and the NDM85/NDM95 can be attributed to the spray drying process (165–185°C) used to produce NDM. Spray drying denatures and damages proteins to varying degrees depending on the inlet air temperature of the drier, leading to the formation of aggregates. Casein proteins are more susceptible to denaturation than whey proteins (as opposed to other conventional heat treatment processes; Fang et al., 2012). The additional pre-heating treatment performed on NDM to obtain NDM95 and NDM85 imparted a notable effect on the protein profiles to cause differences between NDM85 and NDM95, visible on the gels (lanes 3 and 5; Figures 2 and 3). Differences in the surviving background microbiota, variations in the protein structure, or both due to the heat treatments could cause some proteins to become less accessible to *L. helveticus* proteases. Caseins undergo conformational changes at 10, 33, 40, and 78°C at nearly neutral pH (6.75) conditions, but these changes are reversible (Qi et al., 2004). However, any processing treatments before the spray drying of milk concentrates can decrease the digestibility of caseins in an in vitro model simulating the infant's digestive tract (Dupont et al., 2010). Further heat treatments after the spray drying process possibly extended a strain-dependent effect on the digestibility of caseins by *L. helveticus*.

DH and ACE-I Properties of the CFS1 and CFS2

The total protein concentration (mg/mL) in CFS1 and CFS2 was estimated spectrophotometrically and shown in Figures 4A, 4B, 5A, and 5B under the assumption that CFS1 was comprised of whey protein and CFS2 consisted of casein. Because only aromatic residues, Trp and Tyr, absorb light at 278 to 280 nm, those proteins/peptides containing these residues were

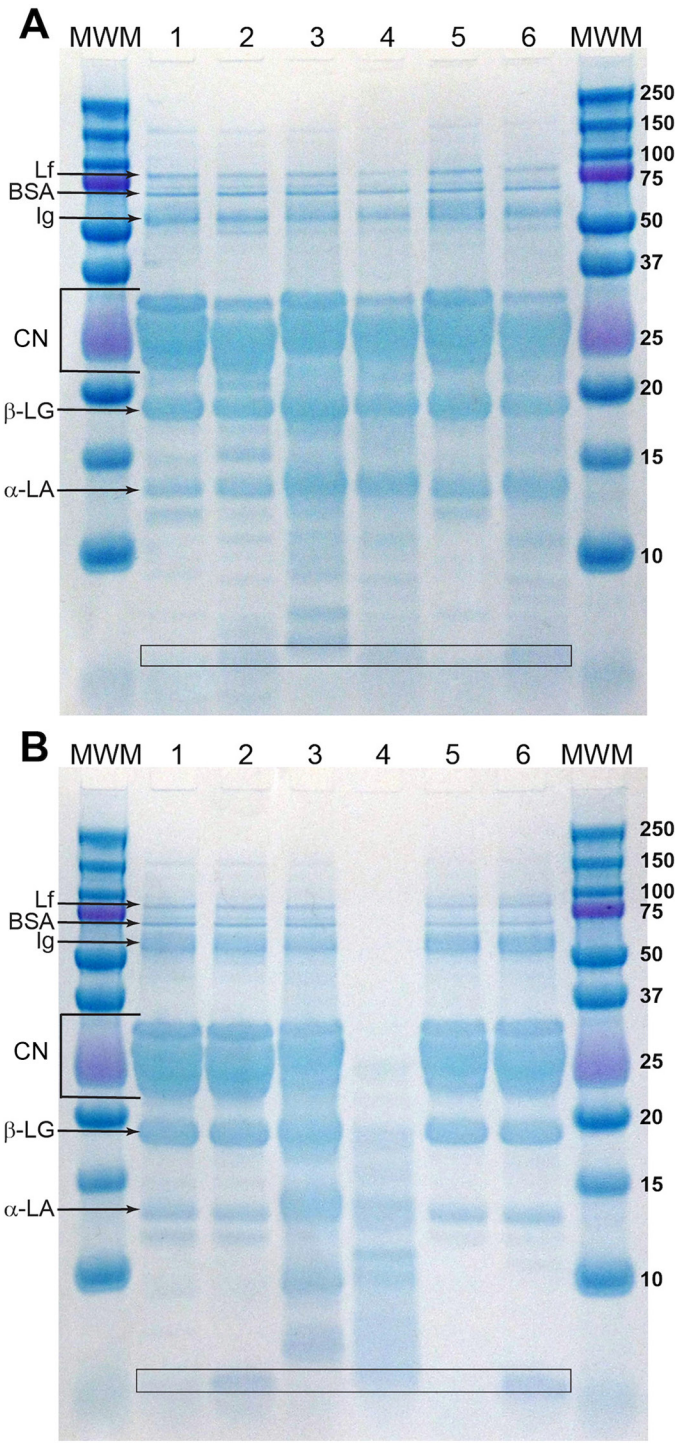


Figure 3. Sodium dodecyl sulfate-PAGE of cell-free supernatant 2 from UHT, NDM95, and NDM85 fermented by *Lactobacillus helveticus* (A) B1929 and (B) ATCC 15009. The lanes were loaded as follows: (1) negative control UHT, (2) *L. helveticus* UHT, (3) negative control NDM95, (4) *L. helveticus* NDM95, (5) negative control NDM85, and (6) *L. helveticus* NDM85. Molecular weight markers (MWM) are also indicated. Unique and yet to be identified bands (due to fermentation) are marked in boxes. Lf = lactoferrin.

included in the protein concentration determination. The molecules lacking these residues, such as small peptides, were mainly precluded. Thus, these concentration estimations possibly underestimated the effective level of proteins/peptides. Nevertheless, these concentrations provided a way to follow and compare the distribution and changes in protein/peptide levels due to fermentation-induced proteolysis.

Notably, the CFS1 of UHT milk from both strains (Figures 4A and 4B) had higher protein concentrations (7.5–8.5 mg/mL) than NDM95 by 3.1 mg/mL ($P < 0.001$) and NDM85 by 3.5 mg/mL ($P < 0.001$). However, their direct acidified counterparts, UHT-AS and NDM-AS, incurred comparable protein concentration (8.5 vs. 7.5 mg/mL) suggesting a significant protein loss occurred in the CFS1 of NDM by fermentation, especially those Trp and Tyr-containing and acid-soluble proteins.

Figures 4C and 4D show the DH in CFS1 by B1929 and ATCC 15009 fermentations. For B1929-fermented milk, the DH of CFS1 (Figure 4C) was 15.0% for UHT compared with 18.0% and 17.8% for NDM (95 and 85, respectively; $P < 0.001$). Additionally, a higher level of hydrolysis occurred in NDM with ATCC 15009 (12.8%; Figure 4D) than in UHT (5.9%; $P < 0.001$). This difference in the level of hydrolysis exerted by B1929 and ATCC 15009 on UHT and NDM warrants further investigation. In addition, pH was negatively correlated with DH ($P < 0.001$, $\rho = -0.86$), as exemplified by the slow-acidifying ATCC 15009 strain in UHT milk. This is at odds with a previous study in which *L. helveticus* L89 degraded casein in pasteurized milk more appreciably under pH controlled (at 6.0) compared with non-pH controlled conditions (Matar et al., 1996). This is reflective of the effect different strains have on nonfat milk proteins.

The ACE-I activity of CFS1 by B1929 and ATCC 15009 was presented in Figures 4E and 4F. Significantly higher ACE-I activity was measured in CFS1 UHT by ATCC 15009 (44.2%) than B1929 (27.9%; $P < 0.001$). However, a study (Wu et al., 2022) was conducted on *L. delbrueckii* fermenting reconstituted NDM subjected to ultra-high-pressure processing and the highest ACE-I activity was obtained after 48- and 72-h incubations. These results suggested that the additional 24-h fermentation with the B1929 strain may have increased its ACE-I activity. On the other hand, other strains of *L. helveticus*, *L. rhamnosus*, or *L. reuteri* did not significantly increase ACE-I activity after 24 h of fermenting milk (Begunova et al., 2021). We also found an increase for NDM95 by ATCC 15009 by 13.4% ($P < 0.001$) compared with UHT. In contrast, the ACE-I of CFS1 NDM85 saw a moderate gain of ~5% by the B1929 fermentation compared with NDM95 ($P < 0.001$).

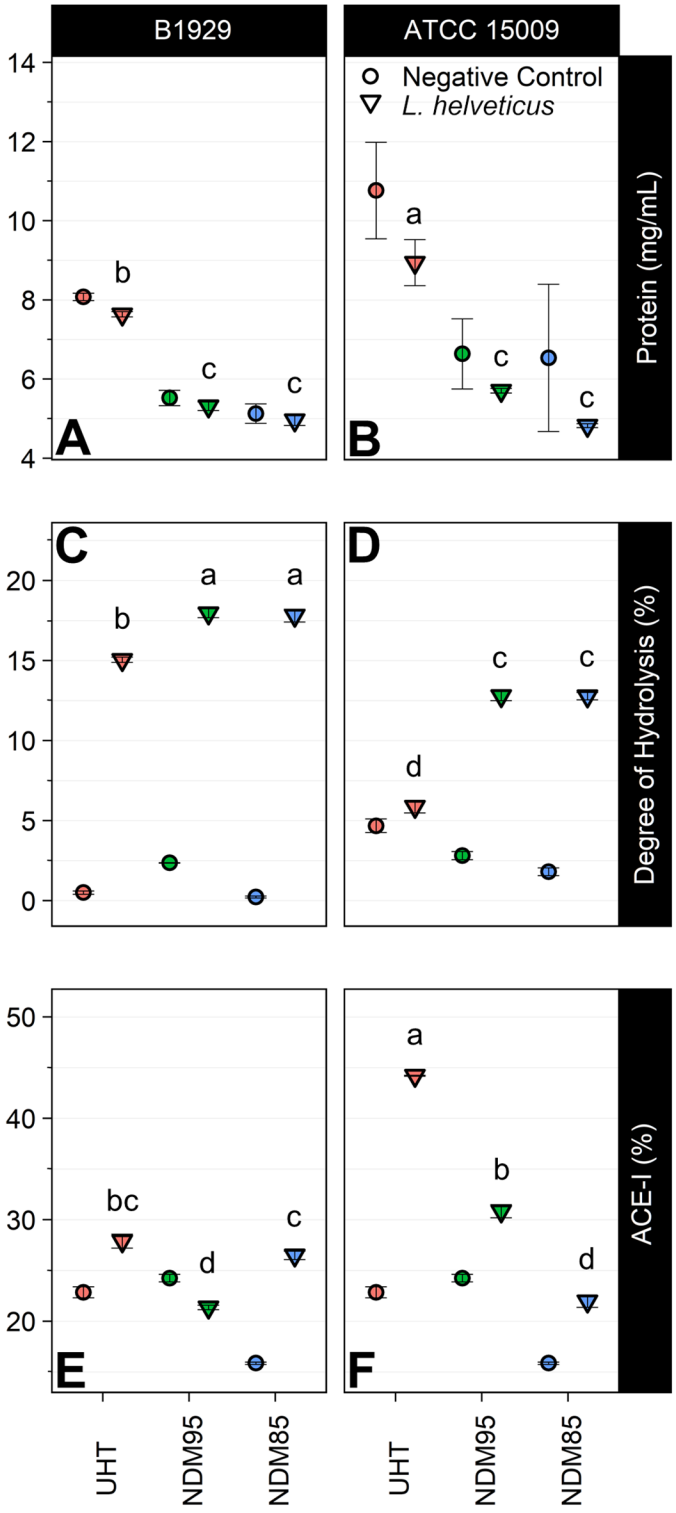


Figure 4. Comparison of protein concentration, degree of hydrolysis, and angiotensin-converting enzyme inhibitory (ACE-I) activity of cell-free supernatant 1. Means (n = 3) and SE bars indicated. Data sharing the same letters were not significantly different ($P > 0.05$) when compared among *Lactobacillus helveticus*-fermented samples. NDM95 = reconstituted milk solution heated to 90 to 95°C for 5 min; NDM85 = reconstituted milk solution heated to 85 to 90°C for 20 min.

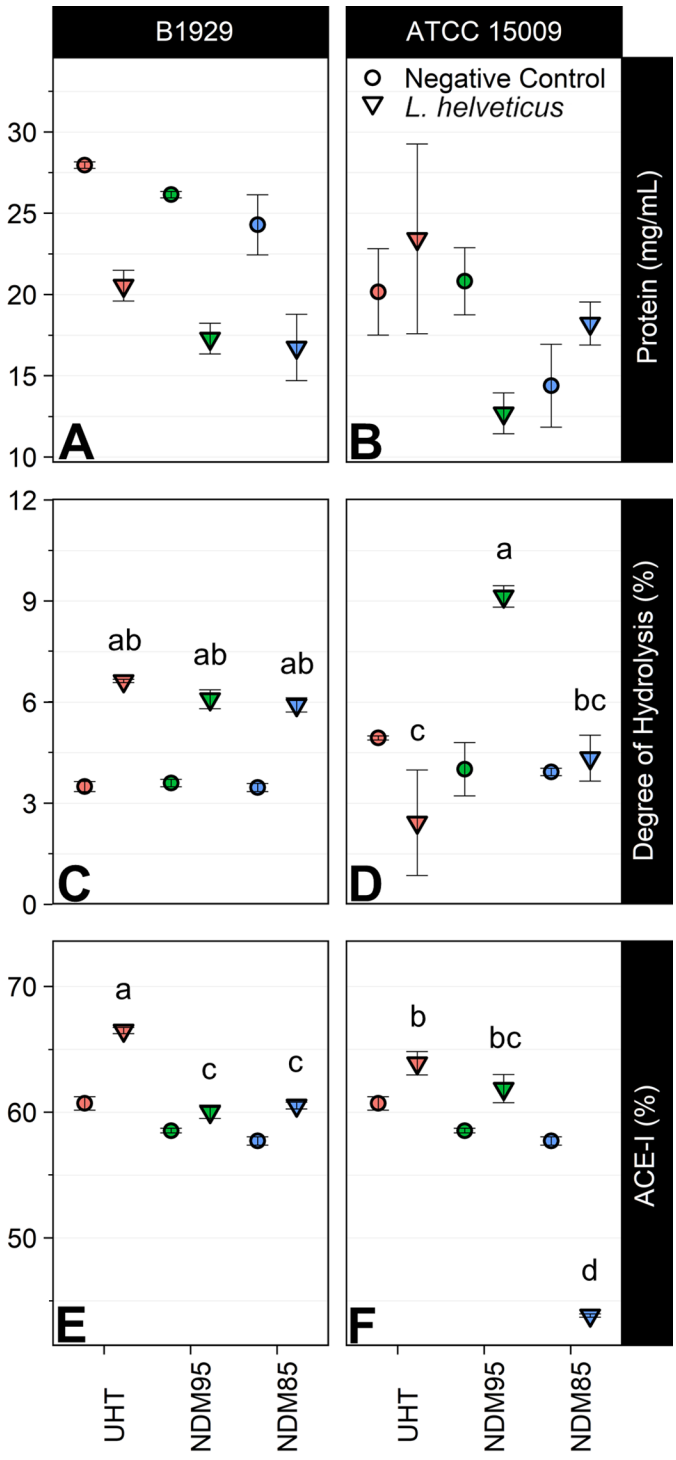


Figure 5. Comparison of protein concentration, degree of hydrolysis, and angiotensin-converting enzyme inhibitory (ACE-I) activity of cell-free supernatant 2. Means (n = 3) and SE bars indicated. Data sharing the same letters were not significantly different ($P > 0.05$) when compared among *Lactobacillus helveticus*-fermented samples. NDM95 = reconstituted milk solution heated to 90 to 95°C for 5 min; NDM85 = reconstituted milk solution heated to 85 to 90°C for 20 min.

It should be noted that almost all CFS1 exhibited elevated ACE-I activity compared with their equivalent acidified fractions, UHT-AS and NDM-AS by 15% and 8.5%, respectively. One exception was B1929, though only by less than 3%, which may mean this strain needed to hydrolyze some peptides with ACE-I activity to survive the conditions of NDM95. The differing behavior between B1929 and ATCC 15009 strains in UHT and NDM will be a subject of continued studies. Despite the differing behavior, a strong negative correlation was detected, determined by Pearson correlation coefficient ($P = 0.001$, $\rho = -0.81$) using these data sets between DH and ACE-I activity. This correlation was in contrast to a study performed with 259 *L. helveticus* strains in pre-heated (95°C for 10 min) 11% reconstituted milk, in which free amino nitrogen was positively correlated with ACE-I activity (Chen et al., 2014). Our findings are unique compared with other studies (Garavand et al., 2022; Loghman et al., 2022) using *L. delbrueckii*, *L. reuteri*, *Lactococcus lactis*, or *Lacticaseibacillus paracasei* to produce ACE-I activity in pre-heated milk. These studies concluded that the degree of proteolysis was associated with a higher level of ACE-I activity. However, the correlations between DH and ACE-I activity from these studies mentioned above were determined by a wide variety of strains in a single milk treatment, making the comparison to the current work challenging.

Figures 5A and 5B show the protein concentrations of CFS2 by B1929 and ATCC 15009 strains, though no significant differences among the fermented samples were observed ($P > 0.05$). Figures 5C and 5D show the DH in CFS2 by B1929 and ATCC 15009 fermentations. The most significant hydrolysis occurred for CFS2 NDM95 by ATCC 15009 (9.1%; Figure 5D), whereas UHT ($P < 0.001$) and NDM85 ($P = 0.005$) remained nearly baseline level around ~2% and ~4%, respectively. This difference in the DH by ATCC 15009 on NDM95 could be due to the pre-heating treatment of NDM at an elevated temperature (95°C). We also detected some decrease in DH in ATCC 15009 in UHT, which may have contributed to the slower growth of this strain compared with its growth in the spray dried and redissolved NDM.

Figures 5E and 5F show the ACE-I activity of CFS2 by B1929 and ATCC 15009. Considerably high ACE-I activity, >60%, was assessed for all CFS2 samples except ATCC 15009 in NDM85, which was 43.8%. Among these, UHT by B1929 had the highest level of ACE-I at 66.5%. This deviation remains to be further examined. Additionally, all CFS2 demonstrated an increased ACE-I activity compared with their respective acidified counterparts, UHT-NS and NDM-NS, 48.5%

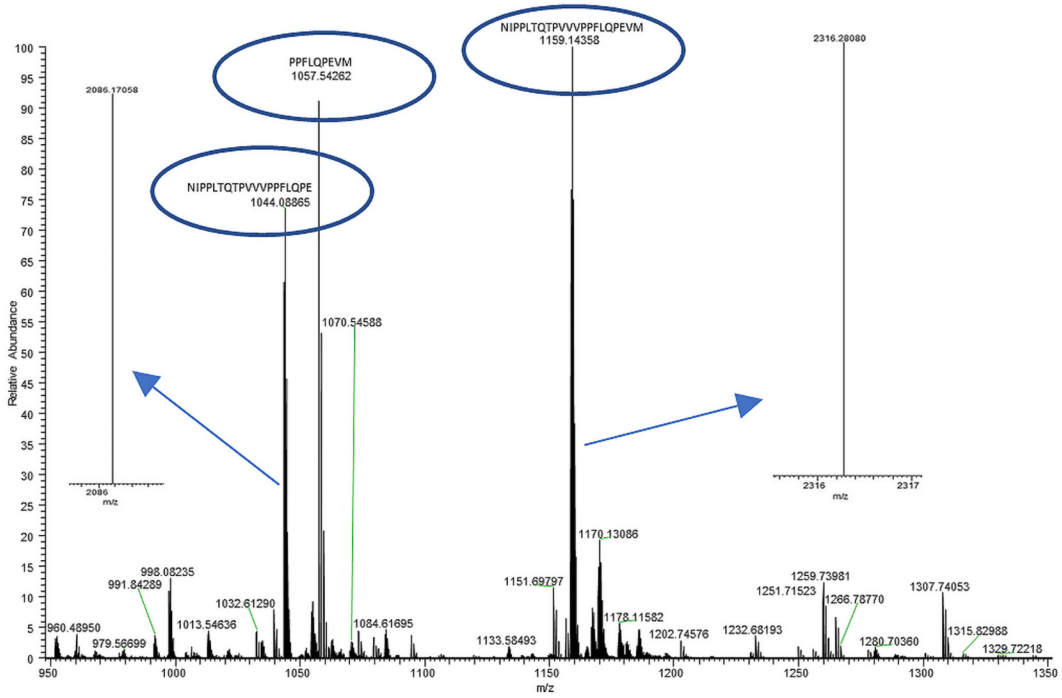


Figure 6. Liquid chromatography-MS chromatograph of cell-free supernatant 1 generated in UHT milk fermented by *Lactobacillus helveticus* B1929.

and 52.3%, respectively. These results showed that fermentation by B1929 and ATCC 15009 significantly improved the ACE-I properties of UHT and NDM of both CFS1 and CFS2 in most cases, as studied in this work, with few exceptions. In the CFS2 of NDM85, the ATCC 15009 dropped the ACE-I activity by 13.3% compared with NDM85-NS. Similar to the decrease in ACE-I of CFS1 NDM95 by B1929, ATCC 15009 had most likely hydrolyzed the peptides with ACE-I activity. Compared with NDM95, protein susceptibility may had been affected by the longer heat treatment, thus making the peptides with ACE-I activity more bioavailable to ATCC 15009 in NDM85, whereas the AA sources without ACE-I activity were more bioavailable in NDM95. Finally, CFS2, mainly casein proteins, showed 33.8% higher ACE-I activity than CFS1.

Peptide Identification

As discussed in section 3.3, a series of small proteins/peptides (<10 kDa) was detected by SDS-PAGE (Figures 2 and 3). Preliminary LC-MS studies were initiated to identify these species. Figure 6 shows the LC-MS chromatograph of the CFS1 fraction generated by UP milk fermentation by *L. helveticus* B12929. Several multiply charged species were detected from the LC-MS runs. Figure 6 represents a spectrum of 2 MH^{2+} species, $m/z = 1{,}044.08865$ and $m/z = 1{,}159.14358$, respectively, and 1 MH^{+} species of 1,057.54262. The spectrum was analyzed by Proteome Discoverer software. The results yielded 17 possible peptides from β-CN with 34% coverage of the protein sequence (shown in Figure 7). Despite of the lack of certainty, the data suggested that β-CN may be involved. This implication of these β-CN fragments has not been reported previously. However, it has been documented (Stepaniak et al., 2001) that the fragments 58 to 72, 193 to 209, and 194 to 209 of β-CN generated by cheese culture showed ACE-I activity. Experiments are currently underway to fully establish the identity of these peptides.

CONCLUSIONS

Results from this study demonstrated that milk processing before fermentation affects the ability of *L. helveticus* strains to grow, resulting in changes in protein compositions, DH, and ACE-I activity. The correlation between the strains used and the pre-conditioning of the medium should be considered when determining the suitability of a LAB strain. The slow-acidifying ATCC 15009 strain produced the highest ACE-I activity in UHT milk in the absence of substantial measurable proteolysis. However, both strains produced a series of β-CN peptides, which may be implicated in the observed ACE-I activity. Further studies are needed to establish this link positively and understand how the

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                              80        90        100       110       160       170                 210       220
Da   P02666    1…71 SLVYPFPGPIPNSLPQNIPPLTQTPVVVPPFLQPEVMGVS 112…159 HQPHQPLPPTVMFPPQSVL 179…206 LYQEPVLGPVRGPFPIIV
1881.1               ---------------------------------------          -------------------          -YQEPVLGPVRGPFPIIV
2089.1               ------------------PPLTQTPVVVPPFLQPEVM---          -------------------          ------------------
1994.1               ---------------------------------------          -------------------          LYQEPVLGPVRGPFPIIV
2086.2               ---------------NIPPLTQTPVVVPPFLQPE-----          -------------------          ------------------
2332.3               ---------------NIPPLTQTPVVVPPFLQPEVM---          -------------------          ------------------
2316.3               ---------------NIPPLTQTPVVVPPFLQPEVM---          -------------------          ------------------
960.5                ------------------------------PFLQPEVM---          -------------------          ------------------
1057.5               ----------------------------PPFLQPEVM---          -------------------          ------------------
1300.7               ----------------------------PPFLQPEVMGVS          -------------------          ------------------
2332.3               ------------------PPLTQTPVVVPPFLQPEVMGVS          -------------------          ------------------
1025.6               --------------------------VVPPFLQPE-----          -------------------          ------------------
1718.0               ---------------------------------------          -------------------          --QEPVLGPVRGPFPIIV
978.6                ------------SLPQNIPPL--------------------          -------------------          ------------------
1300.7               SLVYPFPGPIPN-----------------------------          -------------------          ------------------
1725.9               SLVYPFPGPIPNSLPQ-------------------------          -------------------          ------------------
1156.6               ----------------------------VPPFLQPEVM---          -------------------          ------------------
2150.1               ---------------------------------------          HQPHQPLPPTVMFPPQSVL          ------------------
1017.5               ---------------------------------------          ----------VMFPPQSVL          ------------------
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Figure 7. Amino acid sequence of the peptide series (from cell-free supernatant 1) aligned against the intact β-CN sequence.

behavior of a LAB strain affects its ability to produce specific bioactivity through fermentation.

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